

Arun S. Johnson

MIT PhD ChemE student, environmentalist, and entrepreneur
interested in energy, green electrochemistry, & decarbonization

EDUCATION

Massachusetts Institute of Technology

Prof. Fikile Brushett Lab, *Ph.D. Chemical Engineering*

AUGUST 2023 - PRESENT

University of California, Berkeley

College of Chemistry, *B.S. Chemical Engineering (GPA: 4.0)*

AUGUST 2019 - MAY 2023

EXPERIENCE

Terraform Industries Burbank, CA – Chemical Engineer

MAY 2022 - JULY 2023

- Led the reactor team, charged with designing a Sabatier reactor capable of producing 10 Nm³/hr of pipeline-grade methane from air-captured CO₂ and green hydrogen
- Designed, constructed, and characterized the performance of a Sabatier reactor capable of steady-state hydrogenation of CO₂ at >96% yield with >99% selectivity for methane
- Developed a full set of kinetic, thermodynamic, transport models to inform the design of further reactor iterations

Jeffrey Long Lab UC Berkeley – Undergraduate Researcher

NOVEMBER 2019 - MAY 2023

- Synthesizing and characterizing customizable high-surface area nanomaterials for water purification (PAFs)
- Initiated remote projects in DFT computations (simulated >200 possible binding structures) and 3D printing custom electrochemical cells during lockdown
- Synthesized novel derivatives of N-methyl-D-glucamine to study oxyanion binding mechanisms

PUBLICATIONS

Uliana, A. A; Pezoulas, E. R; Zakaria, N. I; Johnson, A. S; ... Long, J. R. (2024) Removal of Chromium and Arsenic from Water Using Polyol-Functionalized Porous Aromatic Frameworks. ACS. doi:10.1021/jacs.4c05728

Goodman, E. D., Carlson, E. Z., Dietze, E. M., Tahsini, N., Johnson, A., Cargnello, M. (2021). Size-controlled nanocrystals reveal spatial dependence and severity of nanoparticle coalescence and Ostwald ripening in sintering phenomena. Nanoscale. doi:10.1039/d0nr07960j

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[Undergrad Research Summary](#)

[Longform Blog](#) | [Shortform Blog](#)

SKILLS

Programming –

Extensive experience in Matlab, Javascript, Python, Java, DFT with Gaussian 16

Chemical Techniques –

–Competency in FTIR, DLS, H-1 and C-13 NMR, ICP-OES, HRMS, UHPLC, GC, Schlenk line synthesis

–Experience with TEM, SEM, EDX, PXRD, ITC, DSC

Engineering –

3D printing (FDM and SLA), laser cutting, CAD (Fusion 360, Solidworks, Onshape), milling

AWARDS & HONORS

Interact Fellowship (2025)

Fellowship for young technologists

NSF GRFP (2023)

3-year graduate fellowship

Astronaut Scholarship (2022)

Recognizing exceptional scholarship and potential for entrepreneurship in research

Goldwater Scholarship (2022)

UC Berkeley Nominee

Tau Beta Pi (2021)

Engineering honor society

Pioneer Grant (2019)

~\$20k award package for entrepreneurship related to fusion energy research

Emergent Ventures Grant (2019)

\$25k grant for independent research in physics, chemistry, and nuclear fusion

